



SMT49C2410E-43

Axial Fan

About Our Company

Seemtek is derived from 'seem' and 'tech,' suggesting the company's philosophy of making high-tech solutions appear simple and intuitive. Just as the name implies, Seemtek designs fans and motors that blend seamlessly into various environments while delivering advanced performance. These products, like the EC fans, embody the balance of simplicity and technology, providing energy-efficient solutions that adapt effortlessly to different applications, from HVAC systems to industrial use.



- **Mission**

Committed to providing the most tailored fan solutions and products.

- **Vision**

To be the leading brand in external rotor fan technology.

- **Value**

Reliability, Value, Innovation.

We don't just provide products, our goal is to achieve our customers' objectives by offering comprehensive services, from Fan system design to Product delivery and Service. By helping customer engineers solve problems and enhance product performance, Seemtek creates value for its clients. This approach aligns perfectly with our mission: Tailoring Solutions for Fan Industry.



20+
Years
Experience

30+
R&D
Engineers

11
Test
Producers

Our Teamwork

“ At Seemtek's inception, four visionaries united to revolutionize fan technology. ”



River Wu

River Wu, skilled in marketing and managing customer relations, is committed to promoting the products worldwide and fostering strong partnerships with customers.

Michael Zhou

Michael Zhou, a senior fan engineer, leads the team in technical innovation and product design, particularly in electromagnetic technology and fluid design.

David Lee

David Lee is a very experienced production manager, his job is to ensure production lines to operate efficiently, optimizing the manufacturing processes to guarantee products' quality and timely delivery.

Frank Jin

Frankie Jin, a powerful operations expert, optimized production and logistics, doing his best to set up efficient workflows and to establish a state-of-the-art warehouse.

**Their complementary skills
and shared passion driven
Seemtek's success.**

R&D Capability

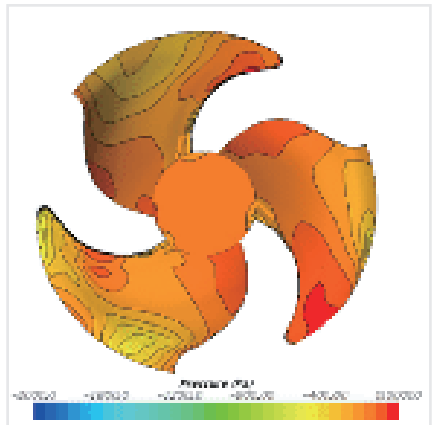
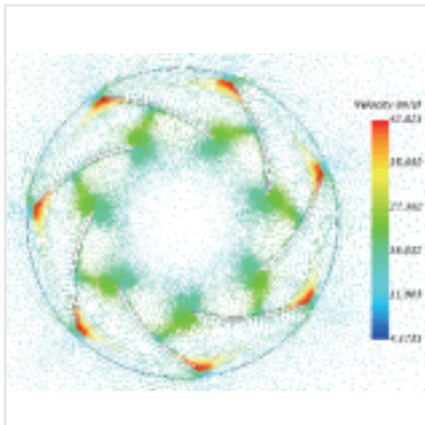
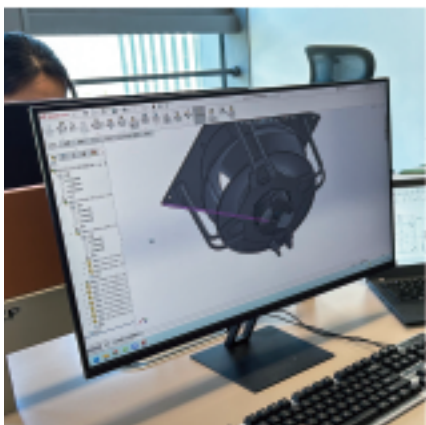
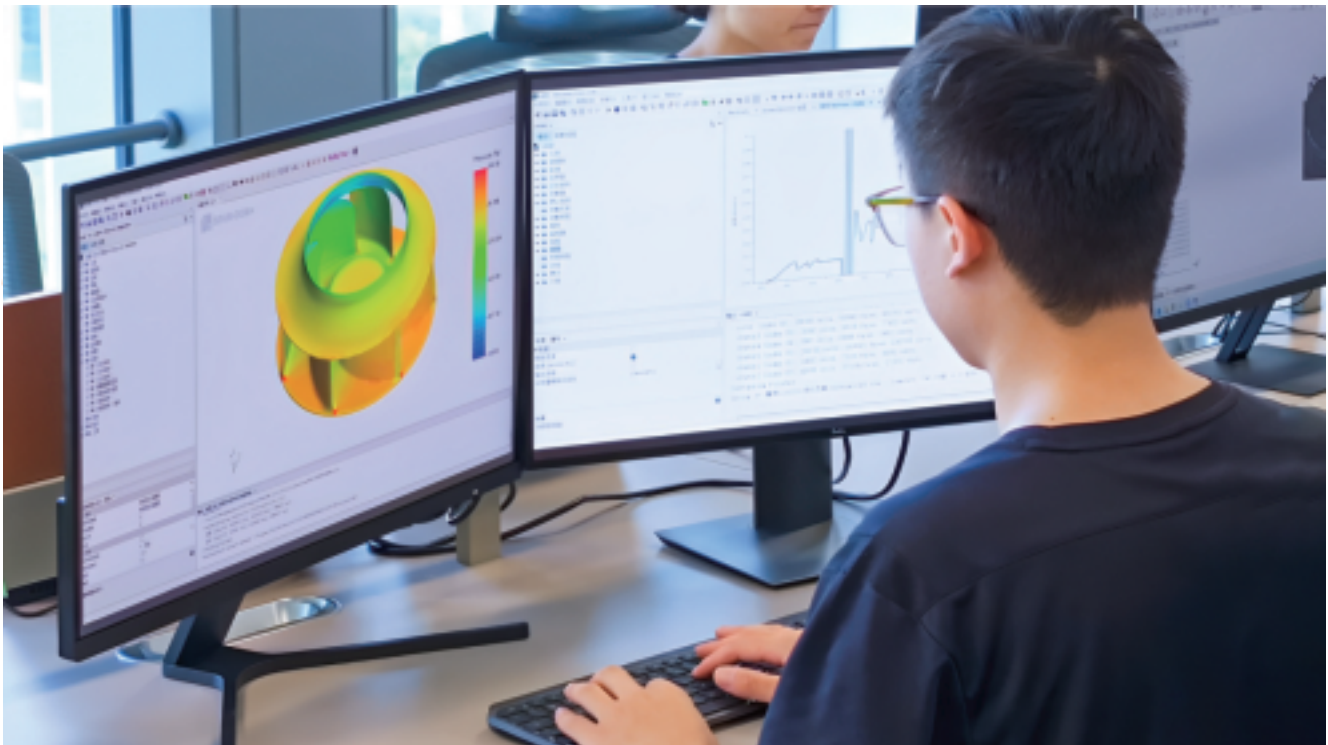
Through 20 years focused on electromagnetic technology and fluid design, Seemtek has always firmly believed that research and innovation are crucial to its competitiveness, which is why it focuses on constantly building and enhancing its R&D division.

Star CCM + CFD

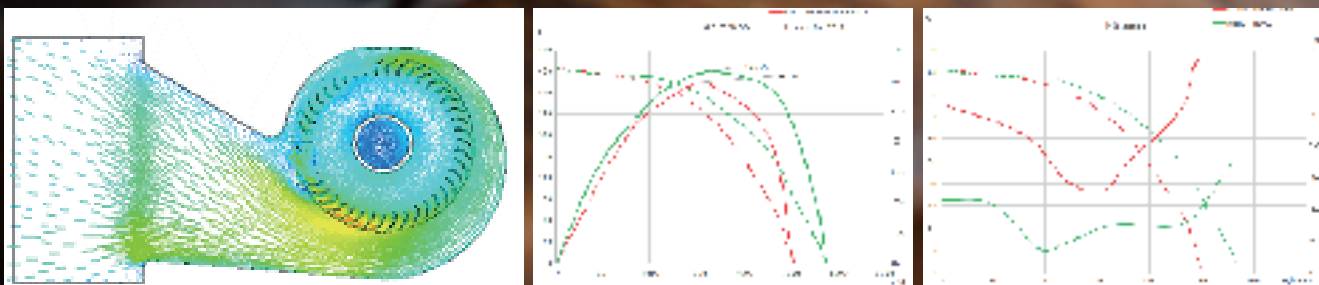
Ansys FEA Altium Designer

Ansoft Maxwell

MSC.patran

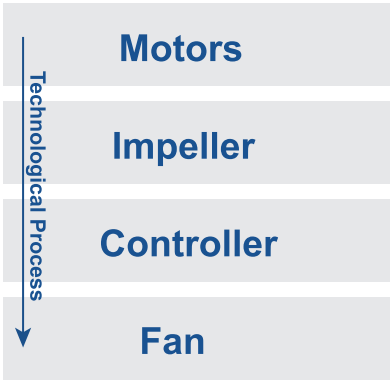


We now have an R&D team of over 30 members. From initial demands to final results, our R&D team always remain by our customers' side, helping them find the most effective solutions to elevate the value of their equipment.



Based on two decades of experience in the fan industry, we have found that appropriate fan selection brings tremendous value to the entire machinery or electromechanical system, and can offer our clients optimal fan selection advice to maximize the effectiveness of their equipment.

Reliability Testing



01

Temperature Shock Test



02

Outdoor Lifespan Test

Our investments in test facilities have always been guided by high market standards, ensure every component employed in Seemtek products is designed and tested internally.

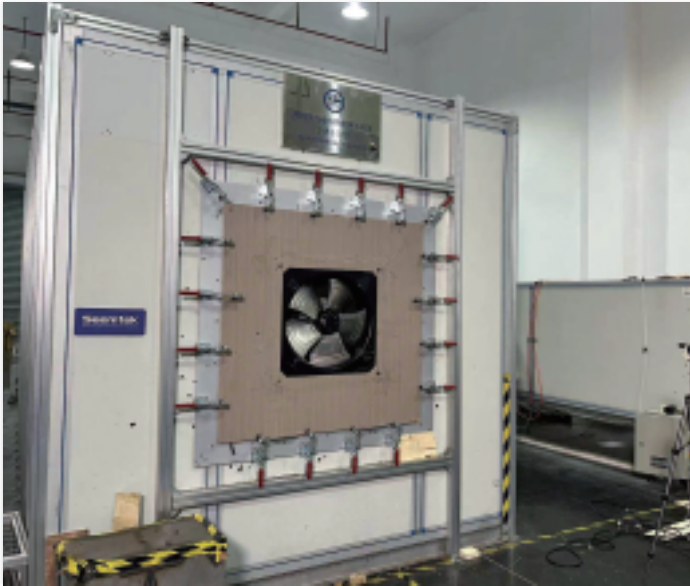
Our comprehensive testing includes airflow tests, temperature tests, and noise tests, lifespan test, totaling 11 test procedures.

03

Salt Mist Test

We ensure that each component passes durability tests.





04

With professional aerodynamics and electromagnetic design, coupled with industry-proven control as well as drive technologies, we ensure that each product achieves excellent performance.

**Airflow
Test**

05

**Noise
Test**



100%
electrical
performance
testing

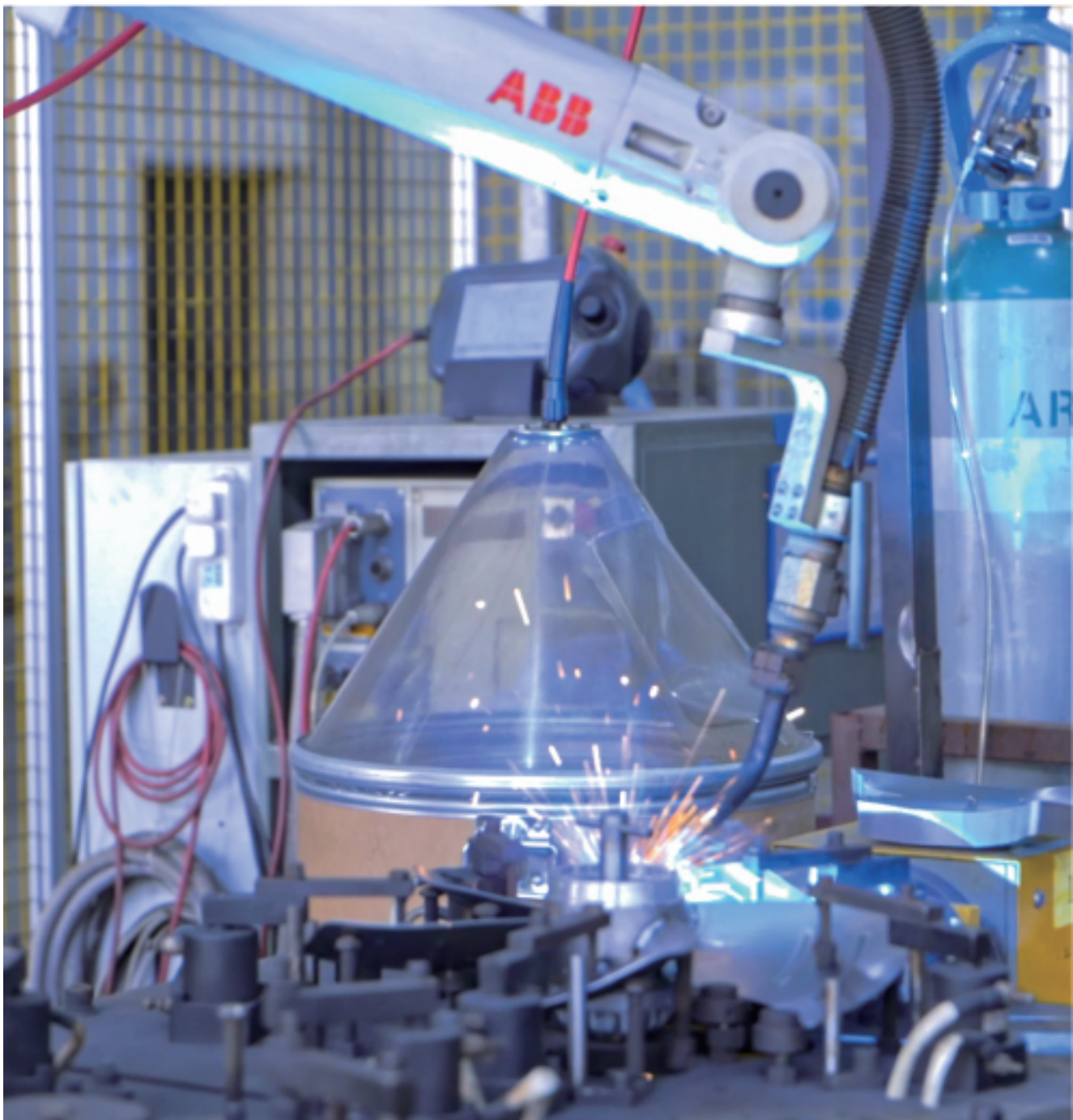


06

Validating the performance of the fan involves comparing the achieved parameters with those set during the design phase to ensure that the fan meets the design requirements.

**High
Temperature Test**

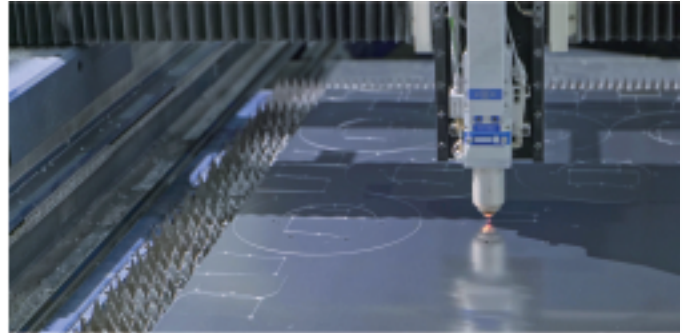
Factory Workshop



Seemtek currently has various fully automated production lines in use, producing millions of high-performance axial fans and centrifugal fans every year. These production lines ensure the highest quality and efficiency, meeting the rigorous demands of our global customers. With the expertise and dedication of our skilled workforce and advanced manufacturing techniques allows us to deliver superior products that set industry standards for performance and reliability.

20+
Years of
Experience
in manufacturing

Millions
of
High-
Performance
Axial
Centrifugal Fans



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Axial Fan

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Principles of EC Fan Model Selection

When selecting an EC fan model, some key factors have to be considered to ensure the chosen fan meet the specific needs of the application while delivering optimal performance and energy savings. The principles below can serve as a guide for making informed decisions in EC fan model selection.



Efficiency

The primary principle guiding the selection of an EC fan model is efficiency. The efficiency of an EC fan is determined by its ability to convert electrical energy into mechanical energy while minimizing losses due to heat dissipation. When selecting an EC fan model, it is important to consider its efficiency rating, typically expressed as a percentage. Higher efficiency ratings indicate a more energy-efficient fan, resulting in reduced power consumption and operating costs.



Airflow and Pressure

Another crucial principle in EC fan model selection is matching the airflow and pressure requirements of the application, known as the operating conditions. In order to achieve a certain airflow, the corresponding system resistance needs to be overcome. The airflow requirement is determined by factors such as the size of the space being ventilated, the desired air exchange rate, and any specific environmental conditions that need to be addressed (e.g., temperature or humidity control). To ensure optimal performance, it is essential to select an EC fan model that can deliver the required airflow while operating within its design parameters. This includes considering the fan's maximum airflow capacity and the ability to adjust the fan speed to meet varying airflow demands.



Noise Level

Noise reduction is an important consideration in many applications, particularly in environments where low noise levels are critical. When selecting an EC fan model, it is important to consider its noise level rating, typically expressed in decibels (dB). Lower noise level ratings indicate a quieter fan, ensuring a more comfortable and productive environment.



Controls and Integration

The ability to integrate the fan into a broader control system is another principle to consider in EC fan model selection. EC fans offer a range of control options, such as variable speed control and the ability to communicate with building management systems. These features allow for precise control over the fan's operation, enabling energy savings through demand-based ventilation. When selecting an EC fan model, it is important to ensure compatibility with the desired control system and assess the available control options to maximize energy efficiency.



Maintenance and Reliability

Lastly, the principle of maintenance and reliability should be considered when selecting an EC fan model. It is essential to choose a fan model that is designed for durability and reliability, reducing the need for frequent maintenance or replacement. Assessing the service life and reliability test that has been done can help ensure a stable operation of the chosen EC fan model.



Part Numbering System

B **3** **E** - **2A** **190** - **S0** - **00**
| | | | | | |
1 **2** **3** **4** **5** **6** **7**

1: Fan Type

B: Backward Curved Centrifugal Fan
K: Centrifugal Fan with Bracket
F: Forward Curved Centrifugal Fan
D: Dual Inlet Blower Fan
S: Single Inlet Blower Fan
A: Axial Fan
G: Axial Fan with Guard Grille
W: Axial Fan with Wall Ring
M: Motor

2: Number of Cores(EC) / Number of Poles(AC)

EC: 1-core and 3-core
AC: 2, 4, 6, 8 and 12 poles

3: Type of Motor

E: EC brushless motor
D: DC brushless motor
A: AC single phase induction motor

4: Input Voltage

12: 12 VDC
24: 24 VDC
48: 48 VDC
85: 85 VDC
1D: 110 VDC
3D: 310 VDC
1A: 115 VAC
2A: 230 VAC
3A: 380 VAC

5: Diameter

190: impeller diameter in mm

6: Key for Electrical Design

7: Key for Mechanical Design

01

Axial

Axial Fan

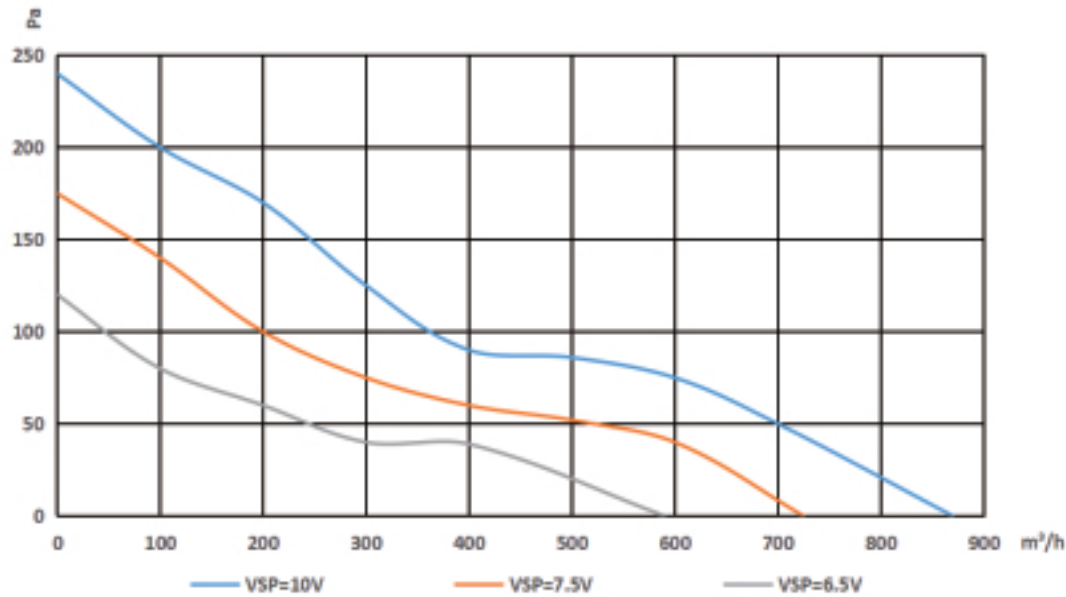


Impeller Material: Plastic
Speed Control: 0-10VDC/PWM
Type of Protection: IP44
Operation Mode: S1
Bearing: Maintenance-free ball bearings
Ambient Temperature: - 25 °C ~ +60 °C
Insulation class: B



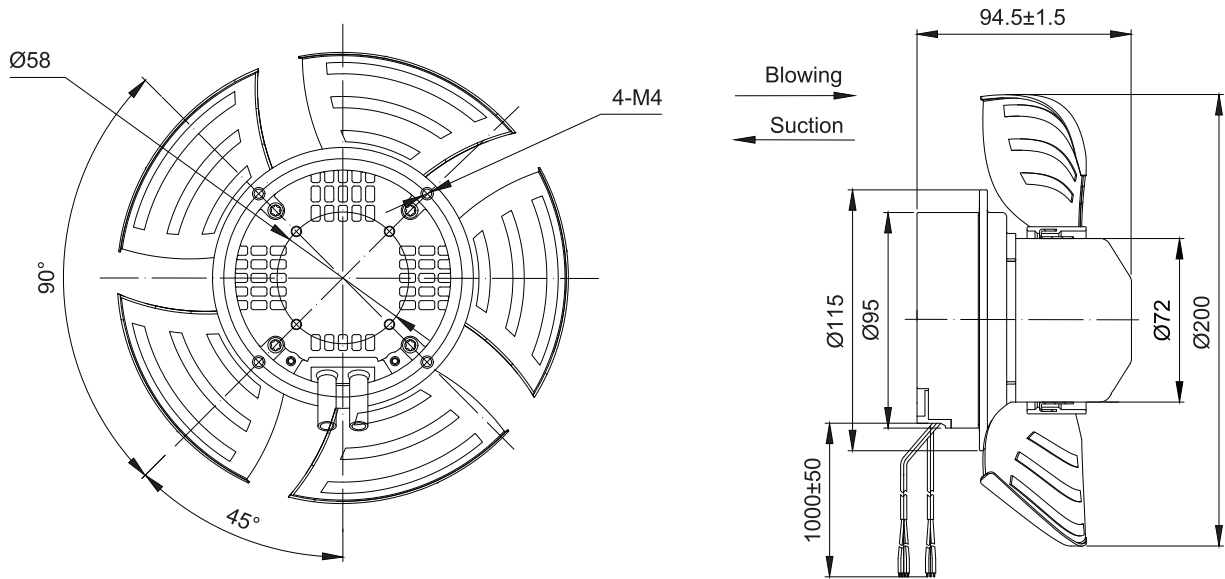
Technical Parameters							
Model No.	Rated Voltage VAC	Frequency Hz	Current A	Input Power W	Speed RPM	Air Flow m³/h	Noise Level dBA
3E-2A200-A7-0	220	50/60	0.60	80	2600	870	54

Performance Curve



Dimensions

Suction	A3E-2A200-A7-01
Blowing	A3E-2A200-A7-02

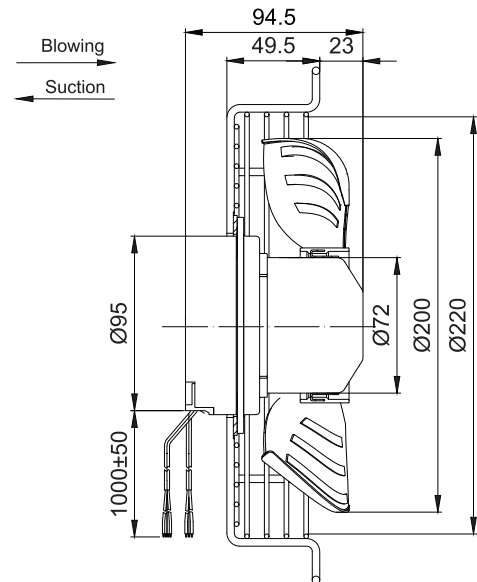
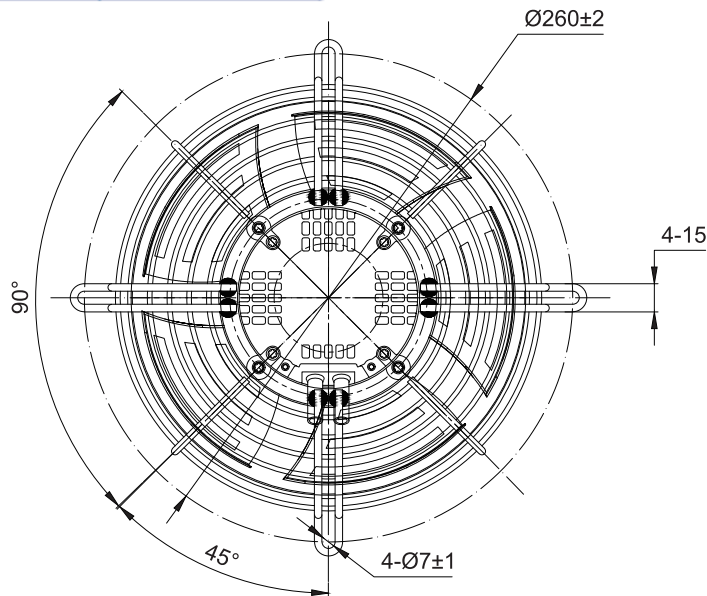


Axial Fan

Dimensions

With wall concave net cover grill

Suction	G3E-2A200-A7-01
Blowing	G3E-2A200-A7-02



Dimensions

With wind guide panel

Suction	W3E-2A200-A7-01
Blowing	W3E-2A200-A7-02

